Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/13/2020
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Kinetics of Electric Field-Driven Phase Transitions in Polarized Colloids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	khusid@njit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	973-596-3316
Organization Name:	New Jersey Institute of Technology		
PI Address 1:	Chemical & Materials Engineering		
PI Address 2:	University Heights		
PI Web Page:			
City:	Newark	**State:**	NJ
Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	08/23/2013	**End Date:**	08/22/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
Contact Email:	walter.m.duval@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/22/2019 per NSSC information (Ed., 8/7/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Emily Ph.D. (NASA Glenn Research Center) Sheehan, Christopher M.S. (ZIN Technologies, Inc.) Loewen, Hartmut Ph.D. (University of Dusseldorf) Telo da Gama, Margarida Ph.D. (University of Lisbon) Yethiraj, Anand Ph.D. (Memorial University of Newfoundland)		
Grant/Contract No.:	NNX13AQ53G		
Performance Goal No.:			
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Task Description:	NOTE: Project continues with grant 80NSSC19K1655, "Advanced Colloids Experiment-Temperature and Gradient Control (ACET11)," period of performance 9/1/2019-8/31/2021. Motivation: Colloidal assembly provides the capability for scalable manufacturing of structured materials and devices for traditional and emerging industries. Although equilibrium structures and thermodynamics of colloids exposed to a field have been well studied, the use of electrically assisted assembly is limited by a lack of understanding of the role of time and spatially varying fields in directing and controlling non-equilibrium processes. Microgravity offers an opportunity to study field-driven phenomena on non-buoyancy-matched colloids by removing masking gravity effects, such as particle sedimentation, convection, and jamming. Objectives: Design and conduct microgravity tests in the International Space Station Advanced Colloids Experiment (ISS ACE-E) facility in long timescales to elucidate fundamental microscopic mechanisms of non-equilibrium phenomena underlying the field-assisted assembly of interacting colloidal particles and suggest novel routes for creating functional materials. The emphasis will be on the field-driven evolution of the mesoscale phase transitions, electro-hydrodynamic instabilities, and nucleation and growth of crystalline structures. The phenomena selected are logical candidates to take advantage of microgravity because terrestrial observations have failed to provide a comprehensive physical understanding of the underlying physics. Methodology: Dielectrophoretic experiments will be carried out on well-characterized colloids under well-defined conditions in terrestrial and microgravity environments. We use electro-hydrodynamic methods and apparatuses for the control and manipulation of particles which were developed in our NASA-supported projects, including parabolic flight tests conducted in 2002 and 2013. Molecular dynamics simulations, coarse-grained and continuum models will be employed to study field-induced colloidal structures. Deliverables: Detailed understanding of electrically driven non-equilibrium phenomena in colloids, apparatuses for the field-assisted colloid assembly, experimental database, and predictive models for the field-induced structure. The research will be integrated with educational and outreach initiatives in space sciences and related technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research has the strong potential to reveal the fundamental microscopic mechanisms of electrically driven phenomena in a suspension of interacting particles and develop a physical explanation for puzzling observations under terrestrial conditions that are not predicted by current theories. The main focus is on the appearance of a cellular structure in a colloid exposed to a spatially uniform field and understanding of the nucleation and growth of crystalline structures in a high-gradient and mechanisms of electro-hydrodynamic instabilities of concentration fronts. The obtained results will facilitate the development of new technologies for the control and manipulation of colloidal processes. Space applications: Electrical technologies can be employed in space to mitigate the effects of reduced gravity by providing an electrical body force. Results of ACE-T-E experiments will be useful for the development of new methods for electro-hydrodynamic control, manipulation, and separation of multiphase fluid flows in various space applications, for example, thermal management systems. Earth applications: The research should bring the understanding of electrically driven colloidal processes to the level useful in the design of apparatuses and processes for a broad range of applications in materials engineering, life science, electronics, and chemical industry. This will stimulate further "spin-offs" of electrical technologies.
Task Progress:	Major Accomplishments for Reporting Period The study was conducted within the scope of the originally-proposed research plan. To specify basic parameters for the design of the ACE-E cell geometry, electrode configuration, voltage/frequency ranges, and procedures of preparing suspensions for the ISS experiments, the New Jersey Institute of Technology (NJIT) team carried out ground-based experiments on suspensions of positively polarized polystyrene/polyaniline (PS/PANI) (1-2 μm) particles and negatively polarized poly-alpha-olefin (PAO) (4-5 μm) particles dispersed in several low-conducting solvents (Mazola corn oil, Envirotemp FR3 and E200 Cargill dielectric fluids, and silicone oil with/without non-ionic surfactant Brij 30). The main objective was to compare structures formed by exposure of suspensions of positively and negatively polarized to an electric field. It is found that only suspensions of negatively polarized poly-alpha-olefin (PAO) particles form cellular structures when exposed to a combination of a sufficiently strong AC field and a relatively weak DC field. Decreasing a particle size increases the ratio between the coulombic DC force and the dipolar AC force exerted on a particle. It is also established that positively polarized PS/PANI particles form columnar patterns when exposed to a sufficiently strong AC field. The contribution of a relatively weak DC field on formation of columns is insignificant. The influence of the size of PAO particles on parameters needed to form cellular structures shows that formation of a cellular structure is dominated by dipolar interaction between particles. The processes of formation of cellular structures following the application of an AC field and then adding a DC field to suspensions of PAO particles increases the ratio between the coulombic DC force and the dipolar AC force causing PAO particles to move onto a positive electrode Development of Requirements for ISS experiments The NJIT team worked in close collaboration with NASA scientists on the design of hardware and experimental procedures conducting for ISS ACE-E experiments. The main focus was on the development of requirements for a high-voltage power supply and an optical system. The PI presented the main results of the proposed ACE-E experiments at the Science Concept Review meeting at NASA Glenn Research Center on November 8, 2017. The Science Review Committee stated that the NJIT team had addressed their concerns and potential problems expressed at the Science Concept Review meeting on November 10, 2015. They stated that the output voltage and frequency ranges of the function generator and amplifier needed for the ISS experiments were consistent with other ACE-E experiments and with commercially available off-the-shelf (COTS) equipment. Requirements for imaging are consistent with several systems available in the ISS. The materials for proposed experiments were fully characterized. The Science Panel also confirmed the adequacy of the proposed test matrix and mission success criteria and stated that the ground-based testing had been taken about as far as possible. We developed the ACE-E Experiment Summary, which includes experimental procedures for conducting microgravity experiments on suspensions of negatively polarized PAO particles and on suspensions of positively polarized PS/PANI particles.

	NOTE: Project continues with grant 80NSSC19K1655, "Advanced Colloids Experiment-Temperature and Gradient Control (ACET11)," period of performance 9/1/2019-8/31/2021.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Abstracts for Journals and Proceedings	Elele E, Lei Q, Khusid B. "Electric field-driven structuring in colloids." Presented at the Concurrent Session: Complex Fluids, 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Abstracts for Journals and Proceedings	Elele E, Lei Q, Khusid B. "Electric field-driven structuring in polarized colloids." Presented at the Concurrent Session: Complex Fluids, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Elele E, Lei Q, Khusid B. "Electric field driven phase transition in a confined suspension of negatively polarized particles." Presented at the 71st Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Atlanta, Georgia, November 18-20, 2018. Bulletin of the American Physical Society. 2018 Nov;63(13):BAPS.2018.DFD.D39.10. https://meetings.aps.org/Meeting/DFD18/Session/D39.10 , Nov-2018
Abstracts for Journals and Proceedings	Lei Q, Elele E, Khusid B. "Electric field driven aggregation of negatively and positively polarized particles in dilute suspensions." Presented at the 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Seattle, WA, November 23-26, 2019. Bulletin of the American Physical Society. 2019 Nov;64(13):A36.00005, http://meetings.aps.org/Meeting/DFD19/Session/A36.5 , Nov-2019
Abstracts for Journals and Proceedings	Lei Q, Elele E, Khusid B. "Electric field driven aggregation of negatively and positively polarized particles in dilute suspensions." Presented at the APS March Meeting 2020, Denver, CO, March 2–6, 2020. Bulletin of the American Physical Society. 2020 Mar;65(1):Abstract:J31.00005. http://meetings.aps.org/Meeting/MAR20/Session/J31.5 , Mar-2020
Articles in Peer-reviewed Journals	Elele E, Shen Y, Tang J, Lei Q, Khusid B. "Single-bubble water boiling on small heater under Earth's and low gravity." npgMicrogravity. 2018 Nov 2;4:21. https://doi.org/10.1038/s41526-018-0055-y ; PMID: 30417085; PMCID: PMC6214906 , Nov-2018